

Natural Language Processing — Test Syllabus

Sample syllabus PDF for testing PrepPilot topic extraction and study-plan generation.

Unit 1 — Introduction to NLP

- Natural Language Processing overview and applications
- Challenges in NLP: ambiguity, context, and language variation
- Text preprocessing: tokenization, normalization, stop-word removal
- Stemming and lemmatization

Unit 2 — Text Representation

- Bag of Words and TF-IDF
- N-grams and feature extraction
- Word embeddings and distributed representations
- Word2Vec: CBOW and Skip-gram

Unit 3 — Language Models

- Statistical language models
- N-gram language models
- Recurrent Neural Networks for language modeling
- LSTM and GRU

Unit 4 — Transformers

- Attention mechanism
- Self-attention and multi-head attention
- Transformer architecture
- BERT and GPT-style models

Unit 5 — NLP Applications

- Sentiment analysis
- Text classification
- Named Entity Recognition
- Question answering and chatbots